University of Glasgow | School of Computing Science

Honours Individual Project Dissertation

Trust Aware Stock Trend Prediction and Investment Strategy Optimisation Using Multimodal Deep Learning

Dylan ShengHong Huang

March 26, 2026

Abstract

This dissertation addresses the gap between accuracy-based metrics and economic value by integrating multi-modal predictions, probability calibration and principled capital allocation into a single end-to-end framework, evaluating whether diverse predictive signals fed into multiple deep-learning architectures (LSTM & GRU variants) trained to predict stock direction and magnitude can yield robust and trustworthy forecasts implemented with risk-aware capital allocation strategies.

The framework incorporates prior model predictions and their historical errors as meta-features, enabling the system to learn when its outputs should be trusted. In addition, a variable-horizon forecasting approach is employed to move beyond traditional fixed-horizon models, allowing the system to capture both short-term fluctuations and long-term trends.

Results show that this approach improves decision stability and reduces excessive turnover compared to uncalibrated baselines, furthermore, risk-aware capital allocation strategies consistently outperform naïve staking strategies, with the hybrid Kelly-Modern Portfolio Theory achieving the strongest risk-adjusted return of TODO: XX% over 100 trading days, TODO: X times higher than the previous baseline set over the same time-frame. These findings lay the groundwork for future research into trust-aware financial prediction systems that prioritises both accuracy and economic value.

Acknowledgements

I would like to express my thanks to Enrique Alejandro González Amador, a former MSc student at the University of Glasgow School of Computing Science, whose dissertation on Adapting Natural Language Processing and Deep Learning Tools to an end-to-end Day Trading System (2025) and, of course, to Prof. Jose Joemon who not only supervised Enrique but myself as well. Without his experience and guidance this project could not have reached the level that it has.

Education Use Consent

I hereby grant my permission for this project to be stored, distributed and shown to other University of Glasgow students and staff for educational purposes. **Please note that you are under no obligation to sign this declaration, but doing so would help future students.**

Signature: Dylan ShengHong Huang Date: 26 March 2026

Contents

1 Introduction

1.1 Motivation

Financial markets are inherently noisy, non-stationary systems in which stock prices are influenced by complex interactions between historical price dynamics and rapidly evolving external information. Despite decades of research, consistently forecasting market behaviour remains challenging, not only due to stochastic price movements but also due to the relationship between predictive signals and future returns varying across time and market regimes. As a result, even small modelling assumptions can significantly impact downstream decision-making.

Recent advances in deep learning have led to measurable improvements in stock movement prediction, particularly through the use of sequence models such as Long Short-Term Memory(LSTM) and Gated Recurrent Unit (GRU) networks, as well as the incorporation of unstructured textual data such as news and social media. However, much of the existing literature evaluates predictive success primarily through accuracy-based metrics, often without sufficient consideration of how model outputs are subsequently used in financial decision-making. In practice, predictive accuracy alone is insufficient, investment decisions depend critically on the reliability of predicted probabilities and the ability to translate forecasts into appropriately sized positions under risk.

In practical investment settings, model outputs are rarely consumed as hard class labels, instead, they are interpreted as probabilities that inform position sizing and risk management. This places importance on both the directional accuracy and the reliability of the prediction. A well-calibrated model produces probabilities estimates whose empirical frequencies align with observed outcomes in practice, whilst poorly calibrated models generate unreliable probabilities that are treated as meaningful inputs to allocation rules leading to over- or under-sized positions, unstable trading behaviour and amplified drawdowns, even when headline accuracy metrics appear strong.

Beyond calibration, a further limitation of many existing approaches is the implicit assumption that a model's predictive reliability is static. In reality, model performance fluctuates over time due to regime shifts, changes in market microstructure and evolving information flows. Models rarely account for their own historical errors when making new predictions and thus may continue to express high confidence even during periods of deteriorating performance. Incorporating information about prior predictions and their realised errors offers a principled way for a system to assess its own reliability and adapt its behaviour accordingly.

Finally, prediction and capital allocation are often treated as separate problems, with models evaluated independently of the investment strategies they enable. This separation obscures the true economic value of predictive improvements and limits the interpretability of reported results Investment performance depends jointly on the quality of predictive signals, the reliability of associated probabilites and the manner in which capital is allocated under uncertainty.

This project is motivated by the need for an integrated framework that unifies multimodal prediction, model self-assessment, probability calibration and principled capital allocation within a single coherent system. By combining heterogeneous data sources with information derived from prior predictions and their realised errors, the framework seeks to improve both the reliability of forecasts and their practical utility under non-stationary market conditions. Through the joint evaluation of statistical performance and economic outcomes, this work aims to bridge the gap between predictive modelling and real-world investment decision-making, demonstrating how certainty-aware learning and risk-sensitive allocation can support more stable and economically meaningful trading strategies in dynamic environments.

1.2 Purpose

The purpose of this project is to design and evaluate an end-to-end, multi-horizon trading system that integrates multimodal predictive signals, probability calibration and risk-aware capital allocation within a single coherent pipeline. Rather than optimising predictive accuracy in isolation, the system is designed to assess the reliability and economic viability of model-driven investment decisions under realistic market conditions.

This work builds upon *Stock Movement Prediction from Tweets and Historical Prices* by **?**, which is used as a baseline for comparison. However, the focus of this project extends beyond next-day prediction and single-model evaluation by exploring variable forecasting horizons, multimodal feature integration and multiple predictive model formulations. Crucially, the value of these predictive signals is assessed through their interaction with capital allocation strategies in a simulated trading environment, rather than through benchmarking metrics alone. To account for non-stationarity and regime shifts, the proposed system incorporates prior predictions and their realised errors as meta-features, enabling the model to capture time-varying predictive reliability.

To this end, the dissertation evaluates multiple deep-learning architectures and systematically examines their integration with several capital allocation strategies, including full Kelly, fractional Kelly, mean–variance portfolio theory (MPT) and a hybrid fractional Kelly-MPT approach. By jointly analysing predictive performance, calibration quality and financial outcomes, this project aims to bridge the gap between statistical forecasting and practical investment decision-making.

1.2.1 Contributions

The following key contributions are made:

- **Stock direction and magnitude prediction formulations**, including binary classification for directional movement, regression for return magnitude and an ordinal classification approach that discretises returns into buckets to estimate expected return.
- **An end-to-end multimodal prediction and investment framework** that integrates historical price data, textual signals and auxiliary features within a single automated pipeline, designed for realistic walk-forward evaluation.
- **A variable-horizon trading system with an early-exit mechanism**, enabling positions to be held for multiple days while allowing premature exit when predicted downside risk increases, thereby improving flexibility relative to fixed-horizon strategies.
- **The explicit incorporation of model self-assessment through meta-features**, using prior predictions and their realised errors to capture time-varying predictive reliability in non-stationary financial markets.
- **A systematic application of probability calibration to deep-learning models**, enabling model confidence scores to be interpreted as meaningful probabilities suitable for downstream risk-sensitive decision-making.
- **A comparative evaluation of capital allocation strategies**, including full Kelly, fractional Kelly, mean–variance portfolio theory (MPT) and a hybrid fractional Kelly–MPT approach, assessed using economic performance measures.

- **An evaluation methodology grounded in economic outcomes**, assessing return, drawdown and risk-adjusted performance alongside traditional accuracy-based predictive metrics.

1.2.2 Novelty

The novelty of this work lies not in proposing new model architectures, but in the integrated treatment of multimodal prediction, uncertainty estimation, model reliability and capital allocation. By embedding these components within a unified framework and evaluating them through economic outcomes, this dissertation advances existing stock prediction research towards more trustworthy and practically deployable system.

1.3 Report Structure

This dissertation is organised into eight chapters.

Chapter 2, *Background and Related Work*, reviews prior research in stock movement prediction and outlines the key theoretical foundations of the project, including LSTM and GRU models, bidirectional variants, ordinal regression, BERT, probability calibration methods (isotonic regression and temperature scaling), the Kelly Criterion, mean-variance portfolio theory (MPT) and the use of meta-features.

Chapter 3, *Analysis and Research Questions*, formalises the problem setting, defines the prediction tasks, discusses constraints such as temporal integrity and non-stationarity as well as presenting the research questions guiding the evaluation.

Chapter 4, *System Design*, describes the architecture of the end-to-end trading framework, including multimodal data integration, model formulations for direction and magnitude prediction, meta-feature incorporation, calibration strategy and the investment simulation design.

Chapter 5, *Implementation*, outlines the practical realisation of the system, covering data preprocessing, feature engineering, model training, calibration, and capital allocation within a walk-forward simulation environment.

Chapter 6, *Evaluation and Results*, reports predictive performance, calibration quality, and economic outcomes such as return, drawdown, and risk-adjusted measures, alongside comparative analyses of model and allocation strategies.

Chapter 7, *Discussion*, interprets the findings in relation to the research questions and reflects on the interaction between multimodal prediction, model reliability, calibration, and capital allocation.

Chapter 8, *Conclusion and Future Work*, summarises the main contributions, discusses limitations, and outlines directions for future research.

Additional material, including extended results and implementation details, is provided in the Appendices.

1.4 Citation styles

- If you are referring to a reference as a noun, then cite it as: "? discusses the role of language in political thought."
- If you are referring implicitly to references, use: "There are many good books on writing (???)."

There is a complete guide on good citation practice by Peter Coxhead available here: `http://www.cs.bham.ac.uk/~pxc/refs/index.html`. If you are unsure about how to cite online sources, please see ?. [1]

1.4.1 Plagiarism warning

WARNING

If you include material from other sources without full and correct attribution, you are commiting plagiarism. The penalties for plagiarism are severe. Quote any included text and cite it correctly. Cite all images, figures, etc. clearly in the caption of the figure.

1.4.2 Quoting text

If you are quoting a long passage, use a `quote` environment:

> If you scribble your thoughts any which way, your readers will surely feel that you care nothing about them. They will mark you down as an egomaniac or a chowderhead –or, worse, they will stop reading you. The most damning revelation you can make about yourself is that you do not know what is interesting and what is not.

(?)

If you are quoting inline, like Simon Peyton-Jones' following remark, use quotation marks "Conveying the intuition is primary, not secondary" (?).

[1] Specifying an online resource like `https://developer.android.com/studio` in a footnote sometimes makes more sense than including it as a formal reference.

2 Background

What did other people do, and how is it relevant to what you want to do?

2.1 Guidance

- Don't give a laundry list of references.
- Tie everything you say to your problem.
- Present an argument.
- Think critically; weigh up the contribution of the background and put it in context.
- **Don't write a tutorial**; provide background and cite references for further information.

3 Analysis/Requirements

What is the problem that you want to solve, and how did you arrive at it?

3.1 Guidance

Make it clear how you derived the constrained form of your problem via a clear and logical process.

The analysis chapter explains the process by which you arrive at a concrete design. In software engineering projects, this will include a statement of the requirement capture process and the derived requirements.

In research projects, it will involve developing a design drawing on the work established in the background, and stating how the space of possible projects was sensibly narrowed down to what you have done.

4 Design

How is this problem to be approached, without reference to specific implementation details?

4.1 Guidance

Design should cover the abstract design in such a way that someone else might be able to do what you did, but with a different language or library or tool. This might include overall system architecture diagrams, user interface designs (wireframes/personas/etc.), protocol specifications, algorithms, data set design choices, among others. Specific languages, technical choices, libraries and such like should not usually appear in the design. These are implementation details.

5 Implementation

What did you do to implement this idea, and what technical achievements did you make?

5.1 Guidance

You can't talk about everything. Cover the high level first, then cover important, relevant or impressive details.

5.2 General guidance for technical writing

These points apply to the whole dissertation, not just this chapter.

5.2.1 Figures

Always refer to figures included, like Figure ??, in the body of the text. Include full, explanatory captions and make sure the figures look good on the page. You may include multiple figures in one float, as in Figure ??, using `subcaption`, which is enabled in the template.

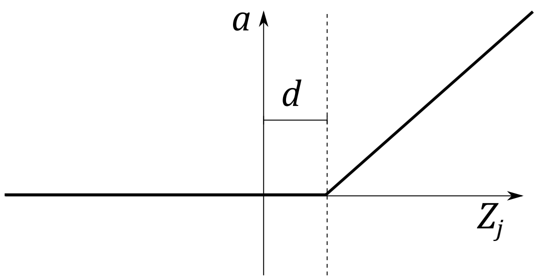
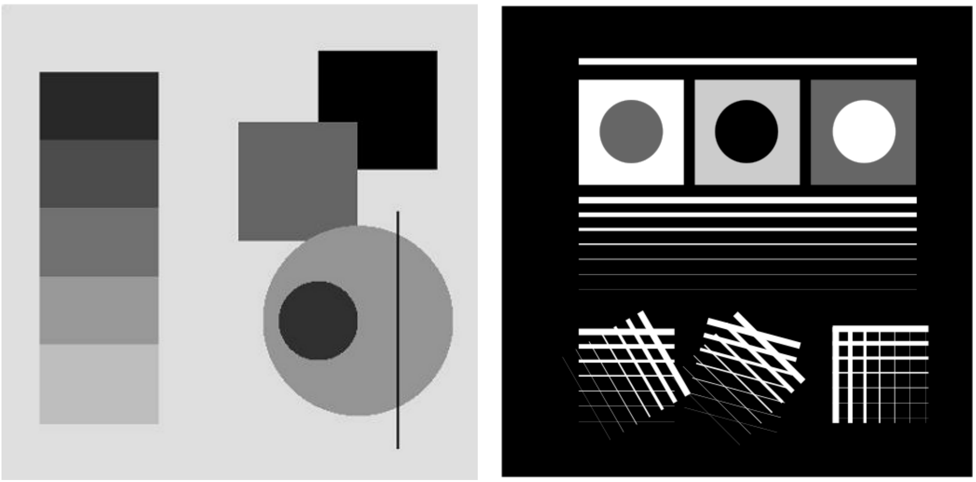


Figure 5.1: *In figure captions, explain what the reader is looking at: "A schematic of the rectifying linear unit, where* a *is the output amplitude,* d *is a configurable dead-zone, and* Z_j *is the input signal", as well as why the reader is looking at this: "It is notable that there is no activation* at all *below 0, which explains our initial results."* ***Use vector image formats (.pdf) where possible****. Size figures appropriately, and do not make them over-large or too small to read.*



(a) *Synthetic image, black on white.* *(b)* *Synthetic image, white on black.*

Figure 5.2: *Synthetic test images for edge detection algorithms. (??) shows various gray levels that require an adaptive algorithm. (??) shows more challenging edge detection tests that have crossing lines. Fusing these into full segments typically requires algorithms like the Hough transform. This is an example of using subfigures, with `subref`s in the caption.*

5.2.2 Equations

Equations should be typeset correctly and precisely. Make sure you get parenthesis sizing correct, and punctuate equations correctly (the comma is important and goes *inside* the equation block). Explain any symbols used clearly if not defined earlier.

For example, we might define:

$$\hat{f}(\xi) = \frac{1}{2}\left[\int_{-\infty}^{\infty} f(x)e^{2\pi i x \xi}\right], \tag{5.1}$$

where $\hat{f}(\xi)$ is the Fourier transform of the time domain signal $f(x)$.

5.2.3 Algorithms

Algorithms can be set using `algorithm2e`, as in Algorithm ??.

Data: $f_X(x)$, a probability density function returing the density at x.
σ a standard deviation specifying the spread of the proposal distribution.
x_0, an initial starting condition.
Result: $s = [x_1, x_2, \ldots, x_n]$, n samples approximately drawn from a distribution with PDF $f_X(x)$.

```
begin
    s ⟵ []
    p ⟵ f_X(x)
    i ⟵ 0
    while i < n do
        x' ⟵ N(x, σ^2)
        p' ⟵ f_X(x')
        a ⟵ p'/p
        r ⟵ U(0, 1)
        if r < a then
            x ⟵ x'
            p ⟵ f_X(x)
            i ⟵ i + 1
            append x to s
        end
    end
end
```

Algorithm 1: The Metropolis-Hastings MCMC algorithm for drawing samples from arbitrary probability distributions, specialised for normal proposal distributions $q(x'|x) = \mathcal{N}(x, \sigma^2)$. The symmetry of the normal distribution means the acceptance rule takes the simplified form.

5.2.4 Tables

If you need to include tables, like Table ??, use a tool like https://www.tablesgenerator.com/ to generate the table as it is extremely tedious otherwise.

5.2.5 Code

Avoid putting large blocks of code in the report (more than a page in one block, for example). Use syntax highlighting if possible, as in Listing ??.

Table 5.1: *The standard table of operators in Python, along with their functional equivalents from the* `operator` *package. Note that table captions go above the table, not below. Do not add additional rules/lines to tables.*

Operation	**Syntax**	**Function**
Addition	`a + b`	`add(a, b)`
Concatenation	`seq1 + seq2`	`concat(seq1, seq2)`
Containment Test	`obj in seq`	`contains(seq, obj)`
Division	`a / b`	`div(a, b)`
Division	`a / b`	`truediv(a, b)`
Division	`a // b`	`floordiv(a, b)`
Bitwise And	`a & b`	`and_(a, b)`
Bitwise Exclusive Or	`a ^b`	`xor(a, b)`
Bitwise Inversion	`~a`	`invert(a)`
Bitwise Or	`a | b`	`or_(a, b)`
Exponentiation	`a ** b`	`pow(a, b)`
Identity	`a is b`	`is_(a, b)`
Identity	`a is not b`	`is_not(a, b)`
Indexed Assignment	`obj[k] = v`	`setitem(obj, k, v)`
Indexed Deletion	`del obj[k]`	`delitem(obj, k)`
Indexing	`obj[k]`	`getitem(obj, k)`
Left Shift	`a <<b`	`lshift(a, b)`
Modulo	`a % b`	`mod(a, b)`
Multiplication	`a * b`	`mul(a, b)`
Negation (Arithmetic)	`- a`	`neg(a)`
Negation (Logical)	`not a`	`not_(a)`
Positive	`+ a`	`pos(a)`
Right Shift	`a >>b`	`rshift(a, b)`
Sequence Repetition	`seq * i`	`repeat(seq, i)`
Slice Assignment	`seq[i:j] = values`	`setitem(seq, slice(i, j), values)`
Slice Deletion	`del seq[i:j]`	`delitem(seq, slice(i, j))`
Slicing	`seq[i:j]`	`getitem(seq, slice(i, j))`
String Formatting	`s % obj`	`mod(s, obj)`
Subtraction	`a - b`	`sub(a, b)`
Truth Test	`obj`	`truth(obj)`
Ordering	`a <b`	`lt(a, b)`
Ordering	`a <= b`	`le(a, b)`

```python
def create_callahan_table(rule="b3s23"):
    """Generate the lookup table for the cells."""
    s_table = np.zeros((16, 16, 16, 16), dtype=np.uint8)
    birth, survive = parse_rule(rule)

    # generate all 16 bit strings
    for iv in range(65536):
        bv = [(iv >> z) & 1 for z in range(16)]
        a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p = bv

        # compute next state of the inner 2x2
        nw = apply_rule(f, a, b, c, e, g, i, j, k)
        ne = apply_rule(g, b, c, d, f, h, j, k, l)
        sw = apply_rule(j, e, f, g, i, k, m, n, o)
        se = apply_rule(k, f, g, h, j, l, n, o, p)

        # compute the index of this 4x4
        nw_code = a | (b << 1) | (e << 2) | (f << 3)
        ne_code = c | (d << 1) | (g << 2) | (h << 3)
        sw_code = i | (j << 1) | (m << 2) | (n << 3)
        se_code = k | (l << 1) | (o << 2) | (p << 3)

        # compute the state for the 2x2
        next_code = nw | (ne << 1) | (sw << 2) | (se << 3)

        # get the 4x4 index, and write into the table
        s_table[nw_code, ne_code, sw_code, se_code] = next_code

    return s_table
```

Listing 5.1: *The algorithm for packing the 3×3 outer-totalistic binary CA successor rule into a $16 \times 16 \times 16 \times 16$ 4 bit lookup table, running an equivalent, notionally 16-state 2×2 CA.*

6 Evaluation

How good is your solution? How well did you solve the general problem, and what evidence do you have to support that?

6.1 Guidance

- Ask specific questions that address the general problem.
- Answer them with precise evidence (graphs, numbers, statistical analysis, qualitative analysis).
- Be fair and be scientific.
- The key thing is to show that you know how to evaluate your work, not that your work is the most amazing product ever.

6.2 Evidence

Make sure you present your evidence well. Use appropriate visualisations, reporting techniques and statistical analysis, as appropriate. The point is not to dump all the data you have but to present an argument well supported by evidence gathered.

If you use numerical evidence, specify reasonable numbers of significant digits; don't state "18.41141% of users were successful" if you only had 20 users. If you average *anything*, present both a measure of central tendency (e.g. mean, median) *and* a measure of spread (e.g. standard deviation, min/max, interquartile range).

You can use `siunitx` to define units, space numbers neatly, and set the precision for the whole LaTeX document.

For example, these numbers will appear with two decimal places: 3.14, 2.72, and this one will appear with reasonable spacing 1 000 000.00.

If you use statistical procedures, make sure you understand the process you are using, and that you check the required assumptions hold in your case.

If you visualise, follow the basic rules, as illustrated in Figure ??:

- Label everything correctly (axis, title, units).
- Caption thoroughly.
- Reference in text.
- **Include appropriate display of uncertainty (e.g. error bars, Box plot)**
- Minimize clutter.

See the file `guide_to_visualising.pdf` for further information and guidance.

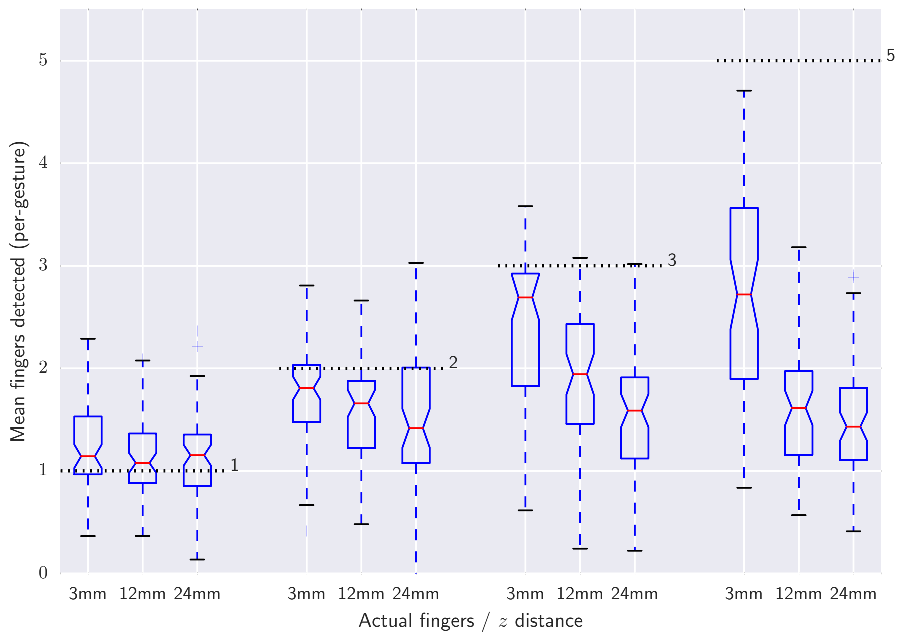


Figure 6.1: *Average number of fingers detected by the touch sensor at different heights above the surface, averaged over all gestures. Dashed lines indicate the true number of fingers present. The Box plots include bootstrapped uncertainty notches for the median. It is clear that the device is biased toward undercounting fingers, particularly at higher z distances.*

7 Conclusion

Summarise the whole project for a lazy reader who didn't read the rest (e.g. a prize-awarding committee). This chapter should be short in most dissertations; maybe one to three pages.

7.1 Guidance

- Summarise briefly and fairly.
- You should be addressing the general problem you introduced in the Introduction.
- Include summary of concrete results ("the new compiler ran 2x faster")
- Indicate what future work could be done, but remember: **you won't get credit for things you haven't done**.

7.2 Summary

Summarise what you did; answer the general questions you asked in the introduction. What did you achieve? Briefly describe what was built and summarise the evaluation results.

7.3 Reflection

Discuss what went well and what didn't and how you would do things differently if you did this project again.

7.4 Future work

Discuss what you would do if you could take this further – where would the interesting directions to go next be? (e.g. you got another year to work on it, or you started a company to work on this, or you pursued a PhD on this topic)

A Appendices

Use separate appendix chapters for groups of ancillary material that support your dissertation. Typical inclusions in the appendices are:

- Copies of ethics approvals (you must include these if you needed to get them)
- Copies of questionnaires etc. used to gather data from subjects. Don't include voluminous data logs; instead submit these electronically alongside your source code.
- Extensive tables or figures that are too bulky to fit in the main body of the report, particularly ones that are repetitive and summarised in the body.
- Outline of the source code (e.g. directory structure), or other architecture documentation like class diagrams.
- User manuals, and any guides to starting/running the software. Your equivalent of `readme.md` should be included.

Don't include your source code in the appendices. It will be submitted separately.